Common Causality of Covariates: a worked example

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This package implements Sunny Karim's CCC procedure. It tests restrictions on untreated covariate coefficients, not parallel trends. HC1 is the default to match the supplied simulation code; it is not cluster robust.

```
> library(ccctest)
> d <- read.csv(system.file("extdata","original.csv",package="ccctest"))
> fit <- ccctest(y_state ~ x, d, "group", "year", "gvar")
> print(fit)

CCC selected specification: region ( holm_region_only )
                W  Q          p     p_holm reject performed used
omnibus 37.741528 23 0.02714581         NA      1         1    1
region  14.387889  5 0.01332446 0.02664892      1         1    1
time     8.704652  5 0.12144054 0.12144054      0         1    1
```

The region-only Holm rejection selects the region specification. Selection is not proof of the maintained model. Both auxiliary fits retain cell intercepts.

```
> tr <- ccc_trends(y_state ~ x, d, "group", "year", "gvar")
> head(tr$series)

        panel time baseline  treated  control       gap
1         raw    1       NA 8.360051 16.89081 -8.530762
2         raw    2       NA 9.100582 18.23836 -9.137777
3 homogeneous    1       NA 8.245915 16.69228 -8.446364
4 homogeneous    2       NA 9.243579 18.19452 -8.950936
5      region    1       NA 8.247648 16.57466 -8.327011
6      region    2       NA 9.250576 17.92598 -8.675405

> head(tr$pairs[, c("pred_st","pred_sb","pred_ct","pred_cb")])

    pred_st  pred_sb  pred_ct  pred_cb
1  4.790557 3.788507 16.42190 14.49074
2  4.790557 3.788507 19.89752 18.39841
3  6.669848 8.517983 16.20648 14.30764
```

```
4  6.669848 8.517983 19.66774 17.95112
5 11.615122 8.045957 16.30996 14.39560
6 11.615122 8.045957 19.77812 18.16599
```

All four predictions in a comparison use the treated target-period empirical covariate distribution. The supplied design has only two common preperiods, so the integrated panel contains one change. Plot with `plot(tr, panels="all")`. For a richer covariate basis use `y_state ~ x + I(x^2) + factor(category)`. The package integrates means of basis columns, not transformations of means. No observation weights, cluster inference, survey design, ATT estimator or confidence bands are provided. A singular requested hypothesis is undetermined. Installation: `R CMD INSTALL r`. Uninstallation: `remove.packages("ccctest")`.